

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n,i,low,high,x;
4      scanf("%d",&n);
5      int a[n];
6      for(i=0;i<n;i++)
7          scanf("%d",&a[i]);
8      scanf("%d",&x);
9      low=0;
10     high=n-1;
11     while(low<=high){
12
13         if(a[low]+a[high]==x){
14             printf("%d\n%d",a[low],a[high]);
15             return 0;
16         }
17         else if(a[low]+a[high] < x)
18             low++;
19         else if(a[low]+a[high] > x)
20             high--;

```

```
19         else
20             high--;
21     }
22     printf("No");
23 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.